Chapter 12. Classification of Surfaces

Fundamental groups of surfaces, homology of surfaces, cutting and pasting, the classification theorem

Contents

12.1 Fundamental groups of surfaces **2**

12.2 Homology of surfaces **3**

12.3 Cutting and pasting **4**

12.4 The classification theorem **4**

12.5 Constructing compact surfaces **6**

12.6 Selected exercises **7**

12.1 Fundamental groups of surfaces

Motivation. Gluing polygon edges in pairs gives every compact surface and a presentation of its π_1.

A **polygonal region** P is the convex hull of points $p_0, \dots, p_{n-1}$ ($n \geq 3$) on a circle in counterclockwise order; its **edges** are the segments $p_{i-1}p_i$ ($p_n = p_0$). An **orientation** of an edge is a choice of initial point; a **labelling** assigns a label to each edge. Pasting edges with the same label by the positive linear maps respecting orientations gives a quotient space X. This is recorded by a **labelling scheme** $w = (a_{i_1})^{\epsilon_1} \cdots (a_{i_n})^{\epsilon_n}$: the k-th edge has label a_{i_k} and is oriented counterclockwise if $\epsilon_k = +1$, clockwise if $\epsilon_k = -1$; several disjoint regions are handled alike. For example, $aba^{-1}b^{-1}$ gives the torus, $abab$ the projective plane (the 2-fold dunce cap), and $aa^{-1}bb^{-1}$ the sphere.

Theorem 12.1 (74.1)

The space X obtained from finitely many polygonal regions by pasting edges according to a labelling scheme is compact Hausdorff.

Proof. Compactness is clear. By Lemma 11.34 (the regions are normal) it suffices that π is closed. For closed C and an edge e, $\pi^{-1}\pi(C) \cap e$ is the union of $C \cap e$ and the $h_i(C \cap e_i)$, $h_i : e_i \to e$ the pasting homeomorphisms; this is compact, hence closed, and $\pi^{-1}\pi(C)$ is the union of C and these finitely many sets. ■

Theorem 12.2 (74.2)

Let X be obtained from a polygonal region P by a scheme $w = (a_{i_1})^{\epsilon_1} \cdots (a_{i_n})^{\epsilon_n}$, with quotient map π. If π maps all vertices to one point x_0 and $a_1, \dots, a_k$ are the distinct labels, then $\pi_1(X, x_0)$ is the quotient of the free group on $\alpha_1, \dots, \alpha_k$ by the least normal subgroup containing $(\alpha_{i_1})^{\epsilon_1} \cdots (\alpha_{i_n})^{\epsilon_n}$.

Proof. $A = \pi(\operatorname{Bd} P)$ is a wedge of k circles, one per label. For each i let f_i be the counterclockwise linear path along an edge labelled a_i, $g_i = \pi \circ f_i$; the $[g_i]$ (or inverses) freely generate $\pi_1(A, x_0)$ (Theorem 11.27). $P \cong B^2$, π maps $\operatorname{Int} P$ bijectively onto $X - A$, and for the loop f once counterclockwise around $\operatorname{Bd} P$, $\pi \circ f = g_{i_1}^{\epsilon_1} * \cdots * g_{i_n}^{\epsilon_n}$ ($g^{-1} = \bar{g}$). Theorem 11.31 gives the result. ■

The **n-fold torus** $T_n = T\# \cdots \#T$ is obtained from a $4n$-gon by the scheme

$$(a_1 b_1 a_1^{-1} b_1^{-1}) \cdots (a_n b_n a_n^{-1} b_n^{-1});$$

for $n = 2$ it is the double torus of Theorem 9.46. The **m-fold projective plane** P_m is obtained from a $2m$-gon by $(a_1 a_1) \cdots (a_m a_m)$ ($m > 1$); set $P_1 = P^2$.

Theorem 12.3 (74.3)

$\pi_1(T_n, x_0)$ is the quotient of the free group on $\alpha_1, \beta_1, \dots, \alpha_n, \beta_n$ by the least normal subgroup containing $[\alpha_1, \beta_1] \cdots [\alpha_n, \beta_n]$, where $[\alpha, \beta] = \alpha\beta\alpha^{-1}\beta^{-1}$.

Proof. By Theorem 12.2 it suffices that all vertices are identified. In a block $aba^{-1}b^{-1}$ on $v_0, \dots, v_4$, a pastes $v_0 \sim v_3$, $v_1 \sim v_2$ and b pastes $v_1 \sim v_4$, $v_2 \sim v_3$; consecutive blocks share a vertex. ■

Theorem 12.4 (74.4)

$\pi_1(P_m, x_0)$ is the quotient of the free group on $\alpha_1, \ldots, \alpha_m$ by the least normal subgroup containing $\alpha_1^2\alpha_2^2 \cdots \alpha_m^2$.

Proof. In a block aa on v_0, v_1, v_2, $v_0 \sim v_1 \sim v_2$; consecutive blocks share a vertex. Apply Theorem 12.2. ■

12.2 Homology of surfaces

Motivation. Abelianizing π_1 gives an easily compared invariant that separates the standard surfaces.

For path-connected X let $\pi_1 = \pi_1(X, x_0)$. For paths α, β from x_0 to x_1, $\hat{\alpha}$ and $\hat{\beta}$ induce the same isomorphism $\pi_1(X, x_0)/[\pi_1, \pi_1] \to \pi_1(X, x_1)/[\pi_1, \pi_1]$: with $g = \alpha * \bar{\beta}$, $\hat{g}([f]) = [g]^{-1}[f][g]$ has the coset of $[f]$. This group, determined up to canonical isomorphism, is the **first homology group** $H_1(X)$; homeomorphisms induce isomorphisms of H_1.

Theorem 12.5 (75.1)

Let $N \lhd F$ and $q : F \to F/N$, $p : F \to F/[F, F]$ the projections. Then p induces an isomorphism $\phi : q(F)/[q(F), q(F)] \to p(F)/p(N)$.

Proof. Let $r : p(F) \to p(F)/p(N)$, $s : q(F) \to q(F)/[q(F), q(F)]$ be projections. $r \circ p$ kills N, inducing $u : q(F) \to p(F)/p(N)$, which kills commutators (abelian target), inducing ϕ. Likewise $s \circ q$ induces $v : p(F) \to q(F)/[q(F), q(F)]$, and $v \circ p = s \circ q$ kills N, so v induces ψ. As $\phi(sq(x)) = rp(x)$ and $\psi(rp(x)) = sq(x)$, they are inverse. ■

Corollary 12.6 (75.2)

Let F be free on $\alpha_1, \ldots, \alpha_n$, N the least normal subgroup containing x, and $G = F/N$. Then $G/[G, G]$ is the quotient of the free abelian group $F/[F, F]$ (basis $p(\alpha_i)$) by the subgroup generated by $p(x)$.

Proof. N is generated by the conjugates of x (Lemma 11.17), all with image $p(x)$, so $p(N) = \langle p(x) \rangle$. Apply Theorem 12.5 and Theorem 11.21. ■

Theorem 12.7 (75.3)

$H_1(T_n)$ is free abelian of rank $2n$.

Proof. By Theorem 12.3 and Corollary 12.6, $H_1(T_n)$ is free abelian on α_i, β_i modulo the image of $[\alpha_1, \beta_1] \cdots [\alpha_n, \beta_n]$, which is 0. ■

Theorem 12.8 (75.4)

The torsion subgroup $T(P_m)$ of $H_1(P_m)$ has order 2, and $H_1(P_m)/T(P_m)$ is free abelian of rank $m - 1$.

Proof. $H_1(P_m)$ is free abelian on $\alpha_1, \ldots, \alpha_m$ modulo $\langle 2\gamma \rangle$, $\gamma = \alpha_1 + \cdots + \alpha_m$. As $\alpha_1, \ldots, \alpha_{m-1}, \gamma$ is another basis, $H_1(P_m) \cong \mathbb{Z}^{m-1} \times \mathbb{Z}/2$. ■

Theorem 12.9 (75.5)

The surfaces S^2, $T_1, T_2, \ldots$, $P_1, P_2, \ldots$ are topologically distinct.

Proof. Their H_1 are 0, $\mathbb{Z}^{2n}$, $\mathbb{Z}^{m-1} \times \mathbb{Z}/2$, pairwise non-isomorphic (by rank, Theorem 11.8, and torsion). ■

12.3 Cutting and pasting

Motivation. Cutting and regluing reduce any scheme to a normal form.

Let P have vertices $p_0, \ldots, p_{n-1}$ and $1 < k < n-1$. Cutting along p_0p_k gives Q_1 (vertices $p_0, \ldots, p_k$) and Q_2 (vertices $p_k, \ldots, p_n = p_0$); translating Q_1 off Q_2, P is homeomorphic to the quotient of $Q_1 \sqcup Q_2$ pasting the two copies of p_0p_k linearly. Conversely, two regions pasted along one edge each form one region (move the vertices of one onto the circle of the other).

Theorem 12.10 (76.1)

Let X be obtained from m regions by the scheme $y_0y_1, w_2, \ldots, w_m$, and let c be a label not appearing in it. If y_0 and y_1 have length at least 2, then X is also obtained from $m+1$ regions by the scheme $y_0c^{-1}, cy_1, w_2, \ldots, w_m$. Conversely, the second scheme gives the same space as the first.

Proof. Cut the first region along the diagonal p_0p_k, k the length of y_0, labelling both copies c (counterclockwise in the second piece, clockwise in the first). A composite of quotient maps is a quotient map, so pasting c first and then the other edges gives the same space. ■

The following **elementary operations** do not change the space; schemes related by them are **equivalent**, and each has an inverse.

(i) *Cut*: replace y_0y_1 by y_0c^{-1} and cy_1 (c new, y_0, y_1 of length ≥ 2).

(ii) *Paste*: replace y_0c^{-1} and cy_1 by y_0y_1 (c appearing nowhere else).

(iii) *Relabel*: replace all occurrences of a label by a new label, or change the sign of all exponents of one label.

(iv) *Permute*: replace y_0y_1 by y_1y_0 (renumber the vertices).

(v) *Flip*: replace $(a_{i_1})^{\epsilon_1} \cdots (a_{i_n})^{\epsilon_n}$ by $(a_{i_n})^{-\epsilon_n} \cdots (a_{i_1})^{-\epsilon_1}$ (reflect the region).

(vi) *Cancel*: replace $y_0aa^{-1}y_1$ by y_0y_1 if a appears nowhere else and y_0, y_1 have length ≥ 2. Cut to y_0ab and $b^{-1}a^{-1}y_1$, merge a, b into one edge c in each (the composite pasting is linear), giving y_0c, $c^{-1}y_1$; paste.

(vii) *Uncancel*: the reverse of (vi).

Example 12.1

The Klein bottle $aba^{-1}b$ is P_2: cut to abc^{-1}, $ca^{-1}b$; permute the first to $bc^{-1}a$; flip the second to $b^{-1}ac^{-1}$, permute to $ac^{-1}b^{-1}$; paste along b: $ac^{-1}c^{-1}a$; permute and relabel to $aabb$.

12.4 The classification theorem

Motivation. Compact connected surfaces are S^2, T_n or P_m, all distinct.

A scheme is **proper** if each label appears exactly twice (preserved by elementary operations). A proper one-region scheme is of **torus type** if each label appears once with each exponent, of **projective type** otherwise. Brackets only group letters; y^{-1} is the formal inverse of a word y.

Lemma 12.11 (77.1)

If $w = [y_0]a[y_1]a[y_2]$ is proper (some y_i possibly empty), then $w \sim aa[y_0y_1^{-1}y_2]$.

Proof.

Step 1. y_0 empty, $w = ay_1ay_2$. If y_1 is empty, done. If y_2 is empty, flip to $a^{-1}y_1^{-1}a^{-1}$, permute and relabel to aay_1^{-1}. Otherwise cut into ay_1c^{-1}, cay_2; flip the first to $cy_1^{-1}a^{-1}$, permute the second to ay_2c; paste along a: $cy_1^{-1}y_2c$; permute and relabel to $aay_1^{-1}y_2$.

Step 2. y_0 nonempty. If y_1, y_2 are empty, permute. Otherwise cut into y_0ac^{-1}, cy_1ay_2; flip and permute the first to $y_0^{-1}ca^{-1}$, permute the second to ay_2cy_1; paste along a: $y_0^{-1}cy_2cy_1$. Permute to $cy_2cy_1y_0^{-1}$; by Step 1, $\sim ccy_2^{-1}y_1y_0^{-1}$; flip to $y_0y_1^{-1}y_2c^{-1}c^{-1}$; permute and relabel to $aay_0y_1^{-1}y_2$. ■

Corollary 12.12 (77.2)

A scheme of projective type is equivalent to one of the same length of the form $(a_1a_1)(a_2a_2)\cdots(a_ka_k)w_1$ with $k \geq 1$ and w_1 empty or of torus type.

Proof. Some label occurs twice with the same exponent, say $+1$; Lemma 12.11 gives $w \sim aaw_1$. If w_1 is not of torus type, write $w = aaz_0bz_1bz_2$ and apply the lemma with $y_0 = aaz_0$: $w \sim bbaaz_0z_1^{-1}z_2$. Each step lengthens the initial block of pairs, so the process ends. ■

One more consequence of the operations:

Step E. *If a does not occur in B, D_1, D_2, then the cyclic scheme $aBa^{-1}D_1D_2$ is equivalent to $aBa^{-1}D_2D_1$.* Take D_1, D_2 nonempty. Permute to $Ba^{-1}D_1D_2a$, cut into $Ba^{-1}D_1c^{-1}$, cD_2a, permute to $D_1c^{-1}Ba^{-1}$, acD_2, paste along a: $D_1c^{-1}BcD_2$; permute and relabel ($c \to a^{-1}$): $aBa^{-1}D_2D_1$. Writing $aBa^{-1}D$ cyclically as $a^{-1}DaB$ and relabelling, the same holds for B: *either block between a and a^{-1} may be cyclically rotated.*

Lemma 12.13 (77.3)

Let $w = w_0w_1$ be proper with w_1 of torus type containing no two adjacent terms with the same label. Then $w \sim w_0w_2$ with w_2 of the same length as w_1 and of the form $aba^{-1}b^{-1}w_3$, w_3 empty or of torus type.

Proof. Let a be a label of w_1 whose two occurrences are closest; they are not adjacent; say a comes first. Let b occur between them, with exponent $+1$ after relabelling; b^{-1} is not between a and a^{-1}. If it follows a^{-1}, $w_1 = y_1aPbQa^{-1}Rb^{-1}y_5$; if it precedes a, exchange a and b (changing exponents). Cyclically $w = aPbQa^{-1}Rb^{-1}S$, $S = y_5w_0y_1$. By Step E: rotate PbQ (between a, a^{-1}) to bQP: $ab(QP)a^{-1}Rb^{-1}S$; rotate $(QP)a^{-1}R$ (between b, b^{-1}) to $a^{-1}RQP$: $aba^{-1}(RQP)b^{-1}S$; rotate $(RQP)b^{-1}S$ (after a^{-1}) to $b^{-1}SRQP$: $aba^{-1}b^{-1}T$, $T = SRQP$. For any $T = T_1T_2$, $aba^{-1}b^{-1}T_1T_2 \sim aba^{-1}b^{-1}T_2T_1$: rotate T_1T_2a (between b^{-1} and b) to T_2aT_1 and repeat the three rotations. With $T_1 = y_5w_0$, $T_2 = y_1RQP$, permuting gives $w_0aba^{-1}b^{-1}w_3$, $w_3 = y_1RQPy_5$ the remaining letters of w_1 with unchanged exponents, so empty or of torus type. ■

Lemma 12.14 (77.4)

If $w = w_0(cc)(aba^{-1}b^{-1})w_1$ is proper, then $w \sim w_0aabbccw_1$.

Proof. Permute to $cc[ab][ba]^{-1}[w_1w_0]$. By Lemma 12.11 backwards ($y_0 = ab$, $y_1 = ba$, $y_2 = w_1w_0$), this is $\sim [ab]c[ba]c[w_1w_0] = [a]b[c]b[acw_1w_0]$, by the lemma $\sim bb[ac^{-1}acw_1w_0] = [bb]a[c^{-1}]a[cw_1w_0]$, and again $\sim aa[bbccw_1w_0]$. Permute. ■

Theorem 12.15 (77.5, classification theorem)

A space obtained from a polygonal region by pasting its edges together in pairs is homeomorphic to S^2, to T_n, or to P_m.

Proof. w is proper of length ≥ 4. We show w is equivalent to one of (1) $aa^{-1}bb^{-1}$ (S^2), (2) $abab$ (P^2), (3) $(a_1a_1)\cdots(a_ma_m)$, $m \geq 2$ (P_m), (4) $(a_1b_1a_1^{-1}b_1^{-1})\cdots(a_nb_na_n^{-1}b_n^{-1})$ (T_n). If w has length > 4 and contains aa^{-1}, cancelling (after permuting so both remaining parts have length ≥ 2) gives a shorter scheme of the same type.

Step 1 (torus type). Length 4: up to permuting and relabelling, $aa^{-1}bb^{-1}$ or $aba^{-1}b^{-1}$. Longer w: induction, assuming no cancellation is possible. Lemma 12.13 gives $w \sim aba^{-1}b^{-1}w_3$, w_3 nonempty of torus type without adjacent pair; apply the lemma with $w_0 = aba^{-1}b^{-1}$, and so on, reaching type (4).

Step 2 (projective type). Length 4: Corollary 12.12 gives $aabb$ (type 3) or aay, $y = b^{\pm1}b^{\mp1}$; writing $y = y_1^{-1}y_2$ with single letters, Lemma 12.11 gives ay_1ay_2, relabelled $abab$. Longer w: Corollary 12.12 gives $(a_1a_1)\cdots(a_ka_k)w_1$. If w_1 is empty, type (3); if it has an adjacent pair, cancel and induct. Otherwise Lemma 12.13 gives $(a_1a_1)\cdots(a_ka_k)aba^{-1}b^{-1}w_2$, and Lemma 12.14 turns this into $(a_1a_1)\cdots(a_{k-1}a_{k-1})aabba_ka_kw_2$, with a shorter torus part. Repeat. ■

12.5 Constructing compact surfaces

Motivation. Triangulations show every compact surface comes from a polygon.

Let X be compact Hausdorff. A **curved triangle** in X is a subspace A with a homeomorphism $h : T \to A$ from a closed triangular region; edges and vertices of A are images of those of T. A **triangulation** of X is a finite collection of curved triangles covering X, two distinct ones meeting in nothing, a common vertex, or a common edge, in the last case with $h_j^{-1} \circ h_i$ linear on the edge of T_i. Every compact surface is triangulable (Radó); we assume this.

Theorem 12.16 (78.1)

A compact triangulable surface is homeomorphic to the space obtained from finitely many disjoint triangular regions by pasting their edges together in pairs.

Proof. Let $A_1, \ldots, A_n$ be a triangulation, $h_i : T_i \to A_i$, T_i disjoint. $h : E = \bigsqcup T_i \to X$ is a closed continuous surjection, hence a quotient map. It suffices that (a) each edge of each A_i is an edge of exactly one other triangle, and (b) if $A_i \cap A_j$ is a vertex v, a chain of triangles with vertex v, consecutive ones sharing an edge, joins A_i to A_j; then h identifies exactly the points identified by the edge pastings.

Step 1 (b). Call triangles with vertex v equivalent if so joined. If there were two classes, let B be the union of one, C of the rest; $B \cap C = \{v\}$. For a neighbourhood W of v missing the triangles not containing v, $W - v$ is the union of the disjoint relatively closed nonempty $(W \cap B) - v$, $(W \cap C) - v$, so disconnected. But v has a disc neighbourhood, hence arbitrarily small W with $W - v$ connected.

Step 2 (at least one other triangle). Let x be interior to an edge of a triangular region T. Its ϵ-neighbourhoods W in T have $W - x$ star-shaped about an interior point, hence simply connected. But if x had a neighbourhood $U \cong B$ (open unit disc, $x \mapsto 0$), then for every neighbourhood $V \subset U$

of x and $B_\epsilon \subset V$, the inclusion $B_\epsilon - 0 \to B - 0$ is homotopic to the homeomorphism $y \mapsto y/\epsilon$, so $\pi_1(V-0) \to \pi_1(B-0) \cong \mathbb{Z}$ is onto and $V-0$ is not simply connected. Now if an edge e of A_i lay in no other triangle, a point interior to e would have a neighbourhood in X inside A_i, and a disc neighbourhood in X would give one in T_i.

Step 3 (at most one). Let $Y \subset \mathbb{R}^3$ be the union of $k \geq 3$ triangles meeting pairwise in a common edge e, $x = 0$ interior to e. The union A of the other edges is k arcs with common end points; it retracts onto three of them (map each extra arc onto one of the three, fixing end points), a theta space with nonabelian π_1 (Example 11.1), so $\pi_1(A)$ is nonabelian. A is a deformation retract of $Y-0$ (radial projection from 0 in each triangle). For a neighbourhood W of 0, choose ϵ with $Y' = \epsilon Y \subset W$; $Y'-0 \to Y-0$ is homotopic to the homeomorphism $y \mapsto y/\epsilon$, so $\pi_1(W-0) \to \pi_1(Y-0)$ is onto and $\pi_1(W-0)$ is nonabelian (base point in Y'). A disc neighbourhood W would have $\pi_1(W-0) \cong \mathbb{Z}$. If e were an edge of three or more triangles, a point interior to e would have a neighbourhood in X inside their union, homeomorphic to such a Y. ■

Theorem 12.17 (78.2)

A compact connected triangulable surface is homeomorphic to a space obtained from one polygonal region by pasting its edges together in pairs.

Proof. By Theorem 12.16, X comes from triangular regions by a proper scheme. While two different regions have edges with the same label, flip one if needed and paste (operation (ii)). If more than one region remained, each would give a nonempty clopen subset of X, contradicting connectedness. ■

By Theorems 12.17, 12.15 and 12.9, every compact connected surface is homeomorphic to exactly one of S^2, T_n, P_m.

12.6 Selected exercises

Motivation. The presentations and H_1 on concrete surfaces, and the list rewritten with P^2 and the Klein bottle.

Exercise 12.1 (§74, Ex. 6–7; nonabelian surface groups)

(a) Show that $\pi_1(T_n)$ is not abelian for $n > 1$.
(b) Show that $\pi_1(P_m)$ is not abelian for $m > 1$.

Solution.

Step 1. (Mapping a presented group.) Let $G = F/N$, F free on γ_i, N the least normal subgroup containing r. For y_i in a group H let $\phi : F \to H$, $\phi(\gamma_i) = y_i$ (Lemma 11.18). If $\phi(r) = e$, then $N \subset \ker \phi$ and ϕ induces $G \to H$. Images of abelian groups are abelian.

Step 2. In S_3, $(1,2)(1,3)$ and $(1,3)(1,2)$ are the two different 3-cycles, so $\langle (1,2), (1,3) \rangle$ is nonabelian.

(a) By Theorem 12.3, $r = [\alpha_1, \beta_1] \cdots [\alpha_n, \beta_n]$. Send $\alpha_1, \beta_1 \mapsto (1,2)$, $\alpha_2, \beta_2 \mapsto (1,3)$, others to e; each $[x,x]$ is trivial, so $r \mapsto e$, and by Steps 1–2 $\pi_1(T_n)$ has a nonabelian image.

(b) By Theorem 12.4, $r = \alpha_1^2 \cdots \alpha_m^2$. Send $\alpha_1 \mapsto (1,2)$, $\alpha_2 \mapsto (1,3)$, others to e; transpositions have order 2, so $r \mapsto e$; conclude as in (a). ■

Exercise 12.2 (§76, Ex. 1; the surface $P^2\#T$)

Let $X = P^2\#T$ be the space obtained from a hexagon by the scheme $aabcb^{-1}c^{-1}$. Calculate $H_1(X)$ and decide which surface of Theorem 12.9 it can be.

Solution.

Step 1. (All vertices go to one point.) Let the kth edge run from p_{k-1} to p_k ($p_6 = p_0$). The a-edges give $p_0 \sim p_1 \sim p_2$; the b-edges ($p_2 \to p_3$, reversed $p_5 \to p_4$) give $p_2 \sim p_5$, $p_3 \sim p_4$; the c-edges ($p_3 \to p_4$, reversed $p_0 \to p_5$) give $p_3 \sim p_0$, $p_4 \sim p_5$.

Step 2. By Theorem 12.2 the relator is $\alpha^2\beta\gamma\beta^{-1}\gamma^{-1}$; by Corollary 12.6, $H_1(X)$ is free abelian on α, β, γ modulo $\langle 2\alpha + \beta + \gamma - \beta - \gamma\rangle = \langle 2\alpha\rangle$:

$$H_1(X) \cong \mathbb{Z}/2 \times \mathbb{Z} \times \mathbb{Z},$$

torsion of order 2, rank 2.

Step 3. Homeomorphic spaces have isomorphic H_1. $H_1(S^2) = 0$ and $H_1(T_n) \cong \mathbb{Z}^{2n}$ are torsion free (Theorem 12.7); $H_1(P_m)$ has torsion of order 2 and rank $m-1$ (Theorem 12.8). So X can only be P_3; Exercise 12.3 shows it is. ■

Exercise 12.3 (§77, Ex. 1(a); another form of the list)

Write K for the Klein bottle, and $T_n\#P^2$, $T_n\#K$ for the spaces given by the schemes $w_n cc$ and $w_n ccdd$, where $w_n = (a_1b_1a_1^{-1}b_1^{-1})\cdots(a_nb_na_n^{-1}b_n^{-1})$. Show that a space obtained by pasting the edges of a polygonal region in pairs is homeomorphic to exactly one of S^2, P^2, K, T_n, $T_n\#P^2$, $T_n\#K$ ($n \geq 1$).

Solution.

Step 1. (A squared letter absorbs a torus block.) For $k \geq 1$ and a word u, applying Lemma 12.14 k times, each time to the pair $(cc)(a_ib_ia_i^{-1}b_i^{-1})$ after the letters already moved,

$$u\,(cc)\,w_k \sim u\,(a_1a_1)(b_1b_1)\cdots(a_ka_k)(b_kb_k)\,(cc).$$

Step 2. ($T_n\#P^2 \cong P_{2n+1}$.) Permute $w_n cc$ to $(cc)w_n$; Step 1 with u empty gives $2n+1$ squares of distinct letters, the scheme of P_{2n+1}.

Step 3. ($T_n\#K \cong P_{2n+2}$.) Permute $w_n ccdd$ to $(cc)(dd)w_n$; Step 1 with $u = cc$ and dd for cc gives $2n+2$ squares: P_{2n+2}.

Step 4. (Existence.) By Theorem 12.15 the space is S^2, some T_n, or some P_m; $P_1 = P^2$, $P_2 \cong K$ (Example 12.1), and for $m \geq 3$, $P_m \cong T_n\#P^2$ ($n = (m-1)/2$, m odd) or $T_n\#K$ ($n = (m-2)/2$, m even) by Steps 2–3.

Step 5. (Uniqueness.) The list is S^2, P_1, P_2, T_n, P_{2n+1}, P_{2n+2} ($n \geq 1$), pairwise distinct standard surfaces, hence non-homeomorphic (Theorem 12.9).

In particular $P^2\#T \cong P_3$, as in Exercise 12.2. ■